MIL-H-81581/1A
1 June 1982
SUPERSEDING
MIL-H-81581/1(AS)
1 December 1969

MILITARY SPECIFICATION SHEET

HOSE ASSEMBLIES, AIR, LOW PRESSURE, HIGHLY FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the hose assembly described herein shall consist of this specification and the latest issue of MIL-H-81581.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a highly flexible low pressure type hose for use with personal survival equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-81581 - Hose Assemblies, Breathing Oxygen and Air; General Specification for.

2.1.2 Other Government documents, drawings, and publications. The following other Government document forms a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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DRAWINGS

NAVAL AIR DEVELOPMENT CENTER

31C1157	Hose Assembly, Air, 7/8 ID; SH3A, S2E, and E2A Aircraft

NAVAL AIR SYSTEMS COMMAND

277AS100	-	Hose Assembly Index, Air, Low Pressure - Highly Flexible

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-H-81581 and as specified herein.

3.2 Qualification. The hoses furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids. Qualification will be in accordance with groups as listed on NASC Index Drawing 277AS100. Qualification approval of any individual hose listed within a group will automatically qualify a manufacturer for all other items listed within the group provided the conditions set forth in MIL-H-81581, paragraph 6.3.1, are complied with.

3.3 Talc. No talc shall be used in the manufacturing process of the hoses furnished under this specification.

3.4 Design and construction. The hose shall be in accordance with the general specification and with the applicable drawing listed on NASC Index Drawing 277AS100.

3.5 Performance. Unless otherwise specified herein, the performance of the hose shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.5.1 Elongation.

3.5.1.1 Procedure I. When tested as specified in the general specification, the increase in length of the hose shall not exceed 12 percent of the original hose length.

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3.5.1.2 Procedure II. When tested as specified in the general specification, the permanent set of the hose shall not exceed 2 percent of the original hose length when measured 2 minutes after removal of load.

3.5.2 Strength.

3.5.2.1 Tensile load. When tested as specified in the general specification, the hose shall not show any indications of failure. Two minutes after the load is removed, the permanent set shall not exceed 1/4 inch.

3.5.2.2 Static load. When tested as specified in the general specification, the hose shall not show a deflection of more than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.5.3 Weight. When tested as specified in the general specification, the hose shall not exceed the following applicable weight:

Hose ID	Weight, maximum
1/2 inch and 5/8 inch	3 ounces per foot
7/8 inch and 1 inch	4 ounces per foot

3.6 Marking. Included with each length of hose, except hose assembly 31C1157, shall be a separate package of 18 white labels marked for identification in accordance with MIL-STD-130. The year and quarter of cure shall also be included. The label material shall be pressure sensitive vinyl cloth tape with a write-on surface. Each label shall be at least 3/4 inch in width and of sufficient length to completely encircle the applicable hose and overlap at least 1/2 inch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose shall consist of the following examinations and tests from the general specification:

Visual examination
Elongation, procedure I
Delamination, procedure I
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure I
Flexibility, procedure I (procedure I (b) under environmental conditions)
Minimum burst pressure, procedure I
Tensile load, procedure I
Static load, procedure I
Low temperature, procedure I
High temperature, procedure I

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Flexibility endurance, procedure I
Ozone-resistance, procedure I
Abrasion, procedure I
Weight, procedure I

4.1.2 First article inspection. The first article inspection of the hose shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article samples shall consist of four hoses of the part number contracted for.

4.1.3 Quality conformance inspection. The quality conformance inspection shall consist of the following tests from the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Elongation, procedure I
Delamination, procedure I
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure I
Flexibility, procedure I
Packaging

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

6.1 Intended use. The hose covered by this detail specification is intended for use in distributing air in the anti-exposure and pressure suit systems.

Custodians:
Air Force 11
Army - AV
Navy - AS

Review Activities:
Air Force - 99

Preparing Activity:
Navy - AS
(Project No. 1660-0444)

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-H-81581/1A Hose Assy, Air, Low Pressure, Highly Flexible

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

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